2.2 The Derivative as a Function; Differentiability

Name _____ Period ____ Date _____

AP Calculus 2.2 · Student sheet

I can…

- find the derivative function f′(x) using the definition.
- connect the graph of f to the graph of f′.
- decide whether a function is differentiable at a number and explain why not.

Vocabulary · fill in the missing word

Derivative function — f′(x) = the _____ as h → 0 of $\frac{f(x+h)-f(x)}{h}$

Differentiable at c — f′(c) _____

Vertical tangent line — a tangent line x = c where the slopes of f _____ ∞ or −∞

Part 1 · Graph the slopes

y = x²: estimate the slope of the tangent line at each x.

x	−2	−1	0	1	2
Slope					

Part 2 · Notes and examples (fill in each note, then work the example)

Derivative function: f′(x) = the limit as h → 0 of _____

Example 1

Use the definition to find f′(x) for f(x) = x² − 3x.

Example 2

Find f′(x) for f(x) = $\sqrt{x}$ and the domain of f′.

Graphs: f′ is positive where f is increasing, negative where f is decreasing, and _____ where f has a horizontal tangent line.

Example 3

f(x) = x² − 4x. Find f′(x). Where is f′ negative, zero, and positive? What does that say about f?

Differentiable means continuous: if f is differentiable at c, then f is _____ at c.

Example 4

$f(x) = x^2$ if $x \le 1$; $2x - 1$ if $x > 1$. Is f differentiable at 1?

Not differentiable at corners, cusps, vertical tangent lines, and _____.

Example 5

Where is each function not differentiable, and why?

a. $f(x) = |x - 2|$

b. $g(x) = \sqrt[3]{x}$

Part 3 · Partner practice

Partner 1

$g(x) = 1 - 2x$ if $x \le 1$; $x - 2$ if $x > 1$. Is g continuous at 1? Differentiable at 1?

Partner 2

(AP® Practice) $f(x) = |x|$. Which of the following are true?

I. f is continuous at 0. II. f is differentiable at 0. III. $f(0) = 0$.

(A) I only (B) III only (C) I and III only (D) I, II, and III